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Optimizing electric vehicle charging patterns and infrastructure for grid decarbonization



Chenlei Liao^{1,5}, Jihao Deng^{2,5}, Xiqun Michael Chen^{1,3}✉ & Quan Yuan^{1,4}✉

Electric vehicles hold decarbonization potential. However, their uncoordinated charging patterns and suboptimal station deployment risk destabilizing power grids. Here we investigate how to enhance grid stability and reduce emissions from mobility-charging-grid perspectives by optimizing charging demand and infrastructure. We present a flexible scheduling strategy to redistribute peak load demand based on spatiotemporal charging and mobility behavior extracted from real-world data spanning 2018 through 2024 in Shanghai, China. Considering projected electric vehicles adoption and population growth, we propose a comprehensive charging station deployment strategy to balance demand and supply, and predict the extra power dispatching emissions caused by charging demand. Results show that by 2035, the strategy could reduce 46.06 thousand tons of cumulative carbon dioxide emissions citywide in household travel. Our findings highlight the importance of integrating charging demand, charging stations, and grid for achieving decarbonization, providing actionable insights for sustainable urban mobility and electrification strategies in large cities.

The combustion of fossil fuels by private vehicles has led to considerable environmental degradation. Statistics reveal that global carbon dioxide (CO₂) emissions from the transportation sector reached nearly 8 Gt CO₂ in 2022, accounting for more than a third of CO₂ emissions from energy end-use sectors¹. Thanks to travel behavior changes and economic development, household travel demand is projected to steadily increase in the coming decades as rising incomes lead to higher vehicle ownership and more frequent travel, resulting in increased greenhouse gas (GHG) emissions². Transportation electrification is regarded as an effective strategy for GHG emissions reduction^{3,4}. Compared to internal combustion engine vehicles (ICEVs), electric vehicles (EVs), including battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs), produce lower GHG emissions as well as air pollutants such as CO₂⁵, nitrogen oxides (NO_x)⁶, and particulate matter (PM)⁷, contributing to improvement in public health^{8,9}. A growing number of countries and regions, including the United States¹⁰, China¹¹, European Union¹², and Japan¹³, have set ambitious automotive electrification targets. They accelerate EV adoption, leading to the rapid growth of global EV market. In 2024, over 58 million EVs were on the road, more than triple the number in 2021¹⁴.

While EVs offer carbon reduction benefits during their operational phase compared to ICEVs^{15–17}, their overall lifecycle emissions may be

higher^{18,19}. In particular, it is often overlooked that the emission reduction benefits associated with EV operations depend on power mix²⁰ and electricity transmission²¹. Emissions could be higher in the regions with high power consumption²², as the electricity transmission network transfers the pollution burden from the usage area to the distant power generation regions²³. Moreover, EV charging can increase peak demand on the grid, potentially leading to grid overload²⁴ and the need for infrastructure upgrades²⁵. Since power generation is the largest source of CO₂ emissions²⁶, the incoordination between power and transportation systems could impede the decarbonization of both sectors²⁷.

To facilitate coordination between power and transportation systems, researchers have proposed strategies, including smart charging²⁸, optimized charging station placement²⁹, and vehicle-to-grid (V2G)³⁰ technologies. These approaches aim to harness the potential of EVs to reduce peak load³¹, lower GHG emissions³², and promote the grid integration of renewable energy³³. However, existing studies are often constrained by data resolution, typically relying on aggregated charging statistics or travel data that lacks explicit charging information, thereby requiring assumptions to estimate drivers' charging decisions^{34,35}. This is an important constraint, as the high heterogeneity and randomness of EV driver behavior^{36,37} mean that even small variations can profoundly influence power demand³⁸.

¹Institute of Intelligent Transportation Systems, College of Civil Engineering and Architecture, Zhejiang University, Hangzhou, China. ²Department of Civil and Mineral Engineering, University of Toronto, Toronto, ON, Canada. ³Zhejiang Provincial Engineering Research Center for Intelligent Transportation, Hangzhou, China. ⁴The Key Laboratory of Road and Traffic Engineering, Ministry of Education, Tongji University, Shanghai, China. ⁵These authors contributed equally: Chenlei Liao, Jihao Deng. ✉e-mail: chenxiqun@zju.edu.cn; quanyuan@tongji.edu.cn

Despite the critical needs for coordination between transportation and power systems, existing research has not sufficiently addressed the coordinated control of individual-level charging behavior, urban charging station placement, and power grid upgrades (Fig. 1a). While some studies have highlighted that the impacts of integrating EVs with grid are closely associated with EV driver behaviors³⁹ and locations of charging stations⁴⁰, they often fall short of providing a long-term perspective. Previous research on charging infrastructure planning has been influenced by paradigms established for conventional vehicle refueling stations⁴¹. These studies have typically prioritized factors such as service coverage⁴², queuing times⁴³, and the maximization of social welfare⁴⁴, all constrained by the limited driving range of EVs⁴⁵. However, the critical interplay and compatibility between vehicles, charging infrastructure, and the grid have been overlooked. These oversights are particularly concerning given that the transition from ICEVs

to EVs is anticipated to span several decades. Collectively, given the substantial variability in individual driver behavior and the influence of factors such as economic and population on the placement of charging stations and power grid upgrades, it is vital to integrate EVs, charging infrastructure, and power grid over an extended time frame for coordination of transportation and power systems, thereby facilitating comprehensive decarbonization of both sectors.

To address this critical integration issue, we present Shanghai, China, as a case study to investigate how to improve power system stability, reduce grid emissions, and facilitate the sustainable electrification of transportation by optimizing EV charging demand and infrastructure deployment. Our analysis is underpinned by a unique dataset with a one-second temporal resolution, affording a detailed view of vehicle mobility patterns and charging behaviors. Specifically, we establish coordination between demand and

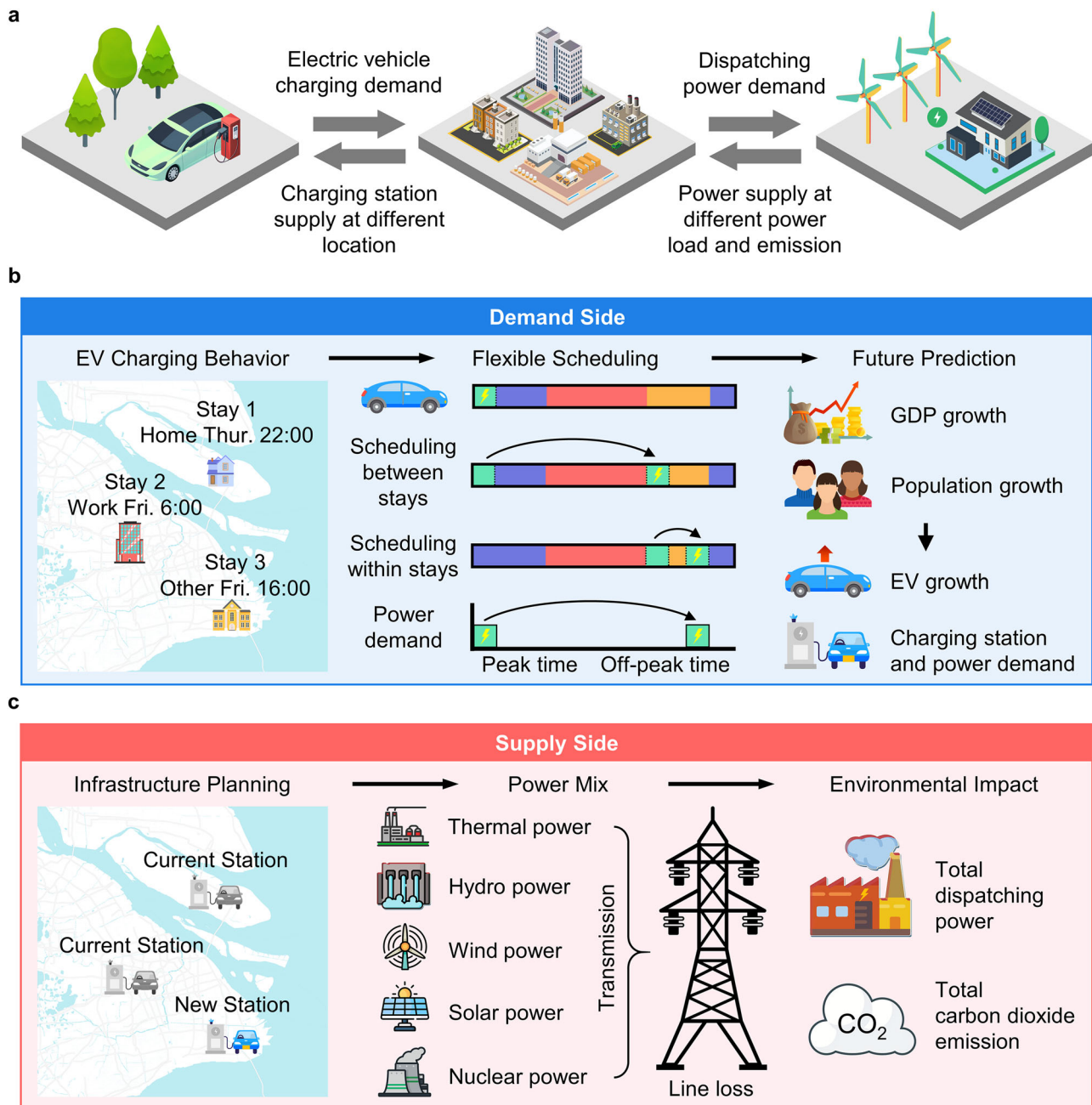


Fig. 1 | Optimizing charging patterns and infrastructure with urban mobility. **a** Supply-demand imbalance among charging patterns, charging infrastructure, and grid. **b** Demand-side optimization and future estimation involving flexible

scheduling based on urban mobility and charging patterns to mitigate load demand during grid peak hours. **c** Supply-side optimization and grid impact analysis based on the post-scheduling demand and power mix.

supply sides by integrating transportation and power systems (Fig. 1b–c). The primary novelty of our study lies in proposing and validating a holistic framework where infrastructure planning is no longer a passive response to demand but an integral component of a dynamic, demand-side management strategy. We demonstrate that by co-optimizing where charging should happen (via scheduling) with where charging can happen (via deployment), cities can achieve far greater grid stability and decarbonization benefits than by addressing these two problems in isolation. First, we analyze spatiotemporal EV charging behavior, identifying temporal peaks and spatial imbalances in charging demand. These findings highlight the need for flexible scheduling strategies to avoid grid overload. Second, according to spatiotemporal mobility patterns, we develop a flexible charging scheduling strategy based on spatiotemporal mobility patterns, which redistributes charging demand and alleviates peak load pressure. For example, suppose an EV arrives home at 10 PM on Thursday and is expected to arrive at a public area by 4 PM on Friday, its charging session at home can be shifted to the public area to avoid grid peak (Fig. 1b). Third, we propose a comprehensive charging station deployment strategy considering population growth, EV adoption rates, and mobility patterns. By optimizing the spatial distribution of charging infrastructure, the strategy supports the effectiveness of scheduling strategies and accommodates future charging demand.

We also analyze the distinct spatiotemporal charging patterns of EVs from the pre-, intra-, and post-pandemic periods, using the real-world data in Shanghai from 2018 to 2024, to assess the effectiveness of the dispatching strategy under varying conditions. Subsequently, we assess the additional annual and cumulative CO₂ emissions resulting from the implementation of these strategies. Finally, a sensitivity analysis is conducted to assess the impacts of driver adoption rate with the scheduling strategies and variations in the scheduling threshold. Our results indicate that coordinating EV charging demand and infrastructure planning can greatly reduce grid stress and emissions, offering actionable insights for sustainable transportation electrification and grid decarbonization in large cities.

Results

Charging demand scheduling with mobility patterns

To assess the charging demand and its impact on the power grid, we first analyze EV charging patterns in 2018 and the load on the power grid in Shanghai (Fig. 2) (see Supplementary Note 1). Our analysis reveals that home charging demand substantially exceeds that of work and other types of charging, with a huge peak occurring at 6 PM through 12 AM. Work charging occurs during 6–10 AM on weekdays. Other charging exhibits a relatively consistent load throughout the day (Fig. 2a). The total charging load generates two distinct peak periods. The higher peak occurs daily at 8 PM, coinciding with Shanghai's lower weekday power load peak at 7–9 PM and higher weekend power load peak at 4–10 PM (Fig. 2b). This overlap may disturb the stability of the power grid. Furthermore, the spatial disequilibrium of charging demand across home, work, and other places may result in a mismatch between charging demand and infrastructure, which restricts the effective off-peak charging strategies implementation (Fig. 2c).

To improve the grid stability and avoid costly power dispatching infrastructure, we propose a flexible charging scheduling strategy aiming at reducing EV charging demand during peak hours (see Methods). Vehicles can transfer charging sessions between available stops within their original route, and charging sessions can be deferred within the stay period. This adaptability in charging schedules leads to a new temporal distribution of electric vehicle charging, effectively reducing the strain on the grid during peak hours. We analyze the stay durations in different regions and the average number of stay sessions for EVs of various types. The tenth percentiles of parking time and stay number distributions were chosen as scheduling thresholds to minimize the impact of the scheduling process (Fig. 2d–e). We divided the study area into 0.5×0.5 km² square cells and allocated EV charging to each cell. We assume that all drivers are subject to scheduling to analyze the optimal impact of the strategy^{21,22}, and extend the results based on the total number of EVs in Shanghai for the relevant year.

Based on the charging load profiles before and after implementing scheduling (Fig. 3a), we observe a maximum charging load reduction of 50.66 MW and a total decrease of 679.86 MWh in peak hours over the seven-day period (41.71% of total demand before scheduling), as the scheduling redistributes the load during the secondary peak to the interval between the two peaks. Home charging accounts for 86.99% of this reduction, contributing to most of the peak load. The shift in charging sessions primarily occurred between home and other (6.78% of home charging shifting to other, and 3.14% shifting to work). This resulted in a decrease in the share of home charging from 78.25% to 72.55%, an increase in other charging from 10.86% to 15.76%, and a slight rise in work charging from 10.89% to 11.68% (Fig. 3b). The total dispatch power demand decreases by 102.42 MWh (7.19% of the pre-scheduling demand) after the scheduling is applied, effectively reducing the dispatch electricity usage. Specifically, the scheduling lowers the regional demand peak from 38.64 MWh to 35.16 MWh with a maximum dispatching load reduction of 4.61 MWh (Fig. 3c). Notably, while almost all charging sessions choose to delay charging times to avoid peak hour, only a small fraction (13.06%) of sessions are subjected to shifting to a different charging location. These results suggest that we require additional public charging infrastructure to implement a scheduling strategy for mitigating power load during peak periods.

Charging station planning for supply-demand balance

Public charging infrastructure deployment is essential for charging scheduling. We collect national and Shanghai GDP from 2000 to 2023, the EV number in Shanghai from 2015 to 2024, and the charging station number in Shanghai from 2018 to 2024 (see Supplementary Note 2–3). We estimate the annual EV growth and the expansion of charging stations in Shanghai from 2025 to 2035 (Fig. 4a–c). The population of Shanghai is estimated using the 2010 China Sixth Census, the 2020 China Seventh Census, along with district-level population statistics for Shanghai from 2018 to 2023 (Fig. 4d–e). Based on our projections, the number of EVs in Shanghai is expected to reach 3.71 million by the end of 2035, the number of public charging stations will be 26.09 thousand, and the total population will be 24.98 million. Notably, the growth of the population and charging infrastructure experienced a slowdown or decline between 2020 and 2022. This trend is attributable to the national state of emergency implemented in China from late 2019 to January 2023 in response to the COVID-19 pandemic.

We aggregate 400 adjacent cells with similar population sizes. The coefficient of determination (R^2) reveals a linear relationship between the square root of the regional population and the number of charging stations within the area (Fig. 4f), which validates the charging infrastructure deployment. Newly established charging stations will be allocated based on the identified aggregated regions, serving to schedule infrastructure demand and accommodate increased demand due to population growth. For scheduling, we analyze the optimal deployment of charging stations under various scheduling strategies and address current deployment gaps. Concerning demographic trends, the allocation of charging infrastructure considers population growth. Based on the 2024 dataset, we examine the charging demand for the year 2035 under optimal scheduling conditions (Fig. 4g). Using estimated charging demand and population growth, we formulate a comprehensive plan for the deployment of public charging facilities from 2019 to 2035 (Fig. 4h). We allocate a maximum of 10% public charging stations to meet scheduling requirements to balance the demand for public and home charging. It ensures that at least 90% of newly deployed stations are reserved to accommodate population growth and support home charging needs (see Methods).

We evaluate scheduling outcomes from 2018 to 2035 based on the deployment of public charging stations (Fig. 5). We segment the timeline into three distinct phases: pre-pandemic (2019 and earlier), intra-pandemic (2020–2022), and post-pandemic (2023 and later). Data from 2018, 2022, and 2024 are used to represent the characteristics of these respective phases. For any given year in our projection, we apply the behavioral patterns and scheduling thresholds from its corresponding phase's baseline year, while

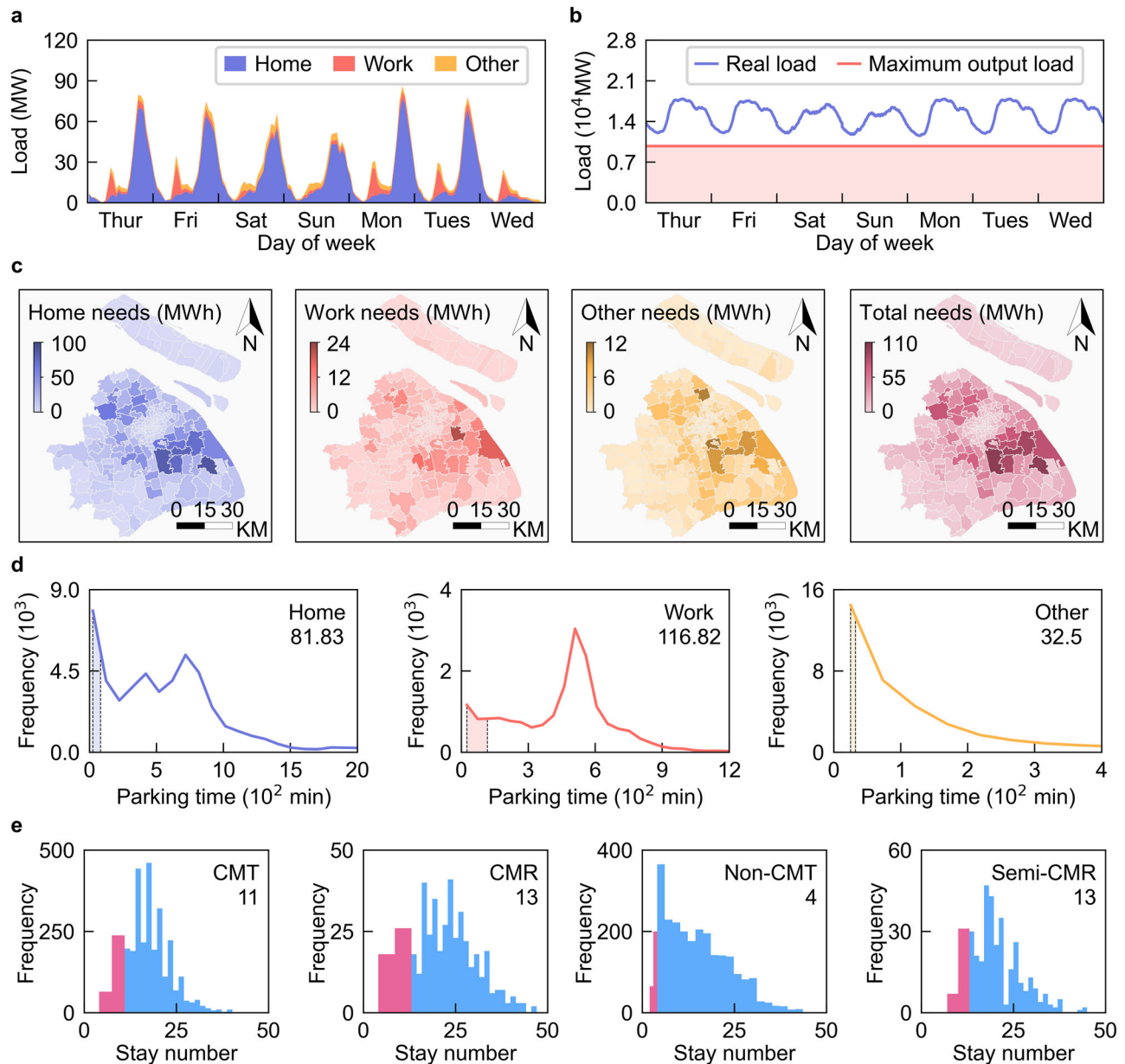


Fig. 2 | Analysis of EV charging patterns in Shanghai. a Temporal distribution of charging demand across different regions. **b** Grid load and maximum power generation capacity of Shanghai. **c** Spatial distribution of charging demand across different regions and total demand. **d** EVs parking time in different regions. The number in the upper-right corner and the shaded area represent the decile (i.e., the

10th percentile) of parking time. **e** The number of stays for different EV types. CMT stands for family-used commuting, CMR for commercially used, Non-CMT for family-used non-commuting, and Semi-CMR for semi-commercially used. The number in the upper-right corner and the plum area represent the decile of parking time.

scaling the total charging demand to match that year's projected EV fleet size (see Supplementary Note 4). This approach allows us to model future impacts based on empirically validated behavioral patterns.

For each period, distinct spatiotemporal charging features are extracted and subsequently used to define the dispatching thresholds (Fig. 5a). Analysis of the pandemic period reveals a great increase in vehicle dwell time, accompanied by a corresponding decrease in trip frequency. The dispatching results demonstrate the strategy's effectiveness across different phases: When compared to the baseline years of 2018, 2020, and 2023, the implementation of our strategy in 2019, 2022, and 2035 result in peak power reductions of 43.46%, 35.78%, and 25.33%, respectively (Fig. 5b). The dispatching strategy's peak reduction performance is more modest in the intra- and post-pandemic scenarios than in the pre-pandemic one, a result of diminished peak-hour charging demand in these periods. A key distinction in the strategy's approach also emerged: Unlike the pre-pandemic model,

which redistributed load to inter-peak shoulders, the later models increasingly deferred charging to the deep overnight valley.

By incorporating the annual EV numbers growth, our 2035 forecast demonstrates substantial savings. By incorporating annual variations in charging demand, we project a substantial reduction in grid stress over a seven-day horizon, including 25.47 GWh decrease in peak power consumption (defined as the total power consumed during grid peak hours), 446.79 MW reduction in average peak load, and a maximum load curtailment of 985.33 MW. Cumulatively, the total peak power consumption savings from 2018 through 2035 are estimated at 10,608.82 GWh. By 2035, charging patterns are expected to evolve, with home and public charging accounting for 40.31% and 47.47%, respectively (Fig. 5c). The post-pandemic era is characterized by a notable increase in "other" charging. This alters the core dispatching logic from a "home-to-other" load shift, typical of earlier periods, to an "other-to-home" shift. The underlying cause is the

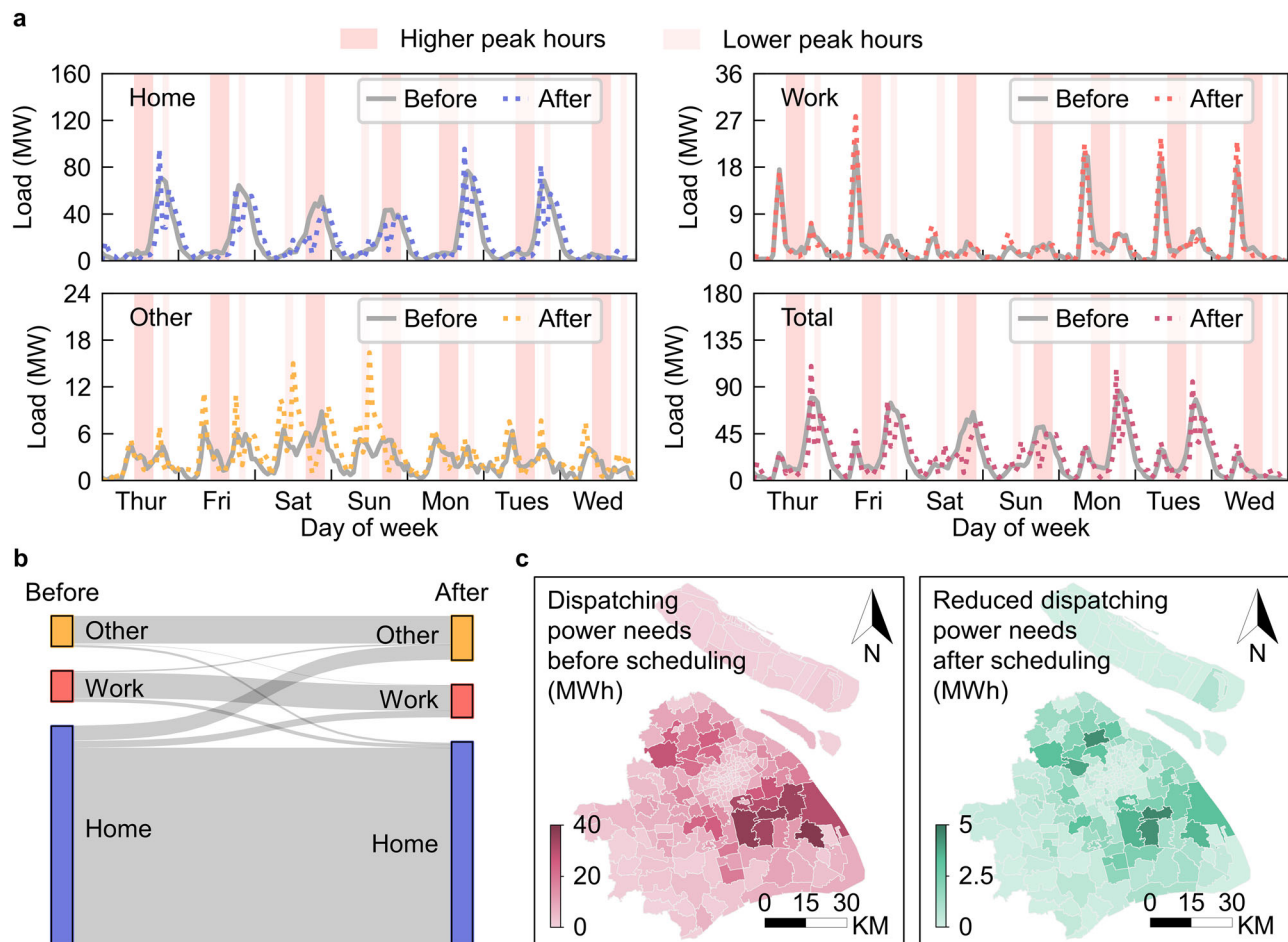


Fig. 3 | Impact of charging scheduling on charging demand. **a** Charging load before and after applying flexible charging scheduling. The red areas denote the grid peak load in Shanghai. The darker red region corresponds to the higher peaks (10:00 AM–4:00 PM on weekdays and 4:00–10:00 PM on weekends), while the lighter red region represents the lower peaks (7:00–9:00 PM on weekdays and 9:30 AM–12:00 PM on weekends). **b** The charging areas before and after applying scheduling. The

blocks on the left and right represent the distribution of charging areas before and after scheduling, with the gray flow indicating the changes in the charging regions resulting from the scheduling application. **c** The original dispatching electricity consumption and the reduced dispatching electricity consumption after the scheduling implementation.

accelerated post-pandemic expansion of Shanghai's EV fleet and public charging network, which created a substantial baseline of "other" charging.

Grid emissions reduction due to charging coordination

Beyond improving grid stability, implementing scheduling strategies and deploying public charging stations may also result in great variations in emissions. To assess the impact of scheduling strategies on power system emissions, we analyze their effects on dispatch power losses (i.e., dispatch electricity $\times$ line losses) and the associated CO₂ emissions. We collected data on the power mix, total electricity consumption, and transmission and line loss rates for the East China sub-grid region, which includes Shanghai (see Supplementary Note 5). Then, we estimate the power system and the impact of scheduling strategies on CO₂ emissions from 2018 to 2035 (Fig. 6). The estimation for the power system suggests that by 2035, the respective shares of hydro, wind, natural gas, coal, nuclear, solar, and biomass in the power mix will be 11.75%, 13.37%, 18.51%, 24.74%, 14.19%, 13.15%, and 3.66% (Fig. 6a). Consequently, the life-cycle CO₂ emission factor is projected to be approximately 291.48 g/kWh, representing a 56.91% reduction from the 2018.

We integrate the power mix into the scheduling results to estimate the average extra CO₂ emissions from dispatch power losses both before and after scheduling (Fig. 6b). Our forecast reveals that the annual extra emissions per vehicle are expected to rise from 13.13 kg in 2018 to 24.55 kg in 2022, before subsequently declining to 14.13 kg by 2035. The initial sharp

increase (2018–2019) was driven by surging charging demand. This was followed by a period of volatility (2020–2022) as pandemic containment policies altered charging behaviors. The long-term trend after 2023 shows a moderate decrease, which is driven by the greater adoption of renewable power and decreased transmission line losses. Although the mitigation potential declined from 9.99% in 2022 to 5.91% in 2023, primarily due to spatiotemporal shifts in charging behaviors after the pandemic, the proposed scheduling strategy remains highly effective in reducing carbon dioxide emissions. By 2035, the implementation of this strategy is projected to reduce annual extra CO₂ emissions per vehicle by 0.99 kg, elevating the mitigation potential to 7.01%.

By multiplying the per-vehicle emissions by the EV number in Shanghai, we estimate the extra CO₂ emissions from dispatch power losses required to meet the annual charging EV demand in the city (Fig. 6c–d). The cumulative extra CO₂ emissions before scheduling reach 665.82 thousand tons from 2018 to 2035. The increased charging demand driven by the rise in EV numbers contributes to the increase in CO₂ emissions. As a result, the extra CO₂ emissions from power scheduling continue to rise, reaching a peak of 52.73 thousand tons in 2034. The extra CO₂ emissions after scheduling in 2035 amount to 48.72 thousand tons, with cumulative emissions reduced to 619.76 thousand tons (Home: 241.36 thousand tons, Work: 75.58 thousand tons, Other: 302.82 thousand tons), representing a 6.92% reduction. Therefore, we can conclude that the extra CO₂ emissions from power dispatching resulting from EV charging in Shanghai will continue to

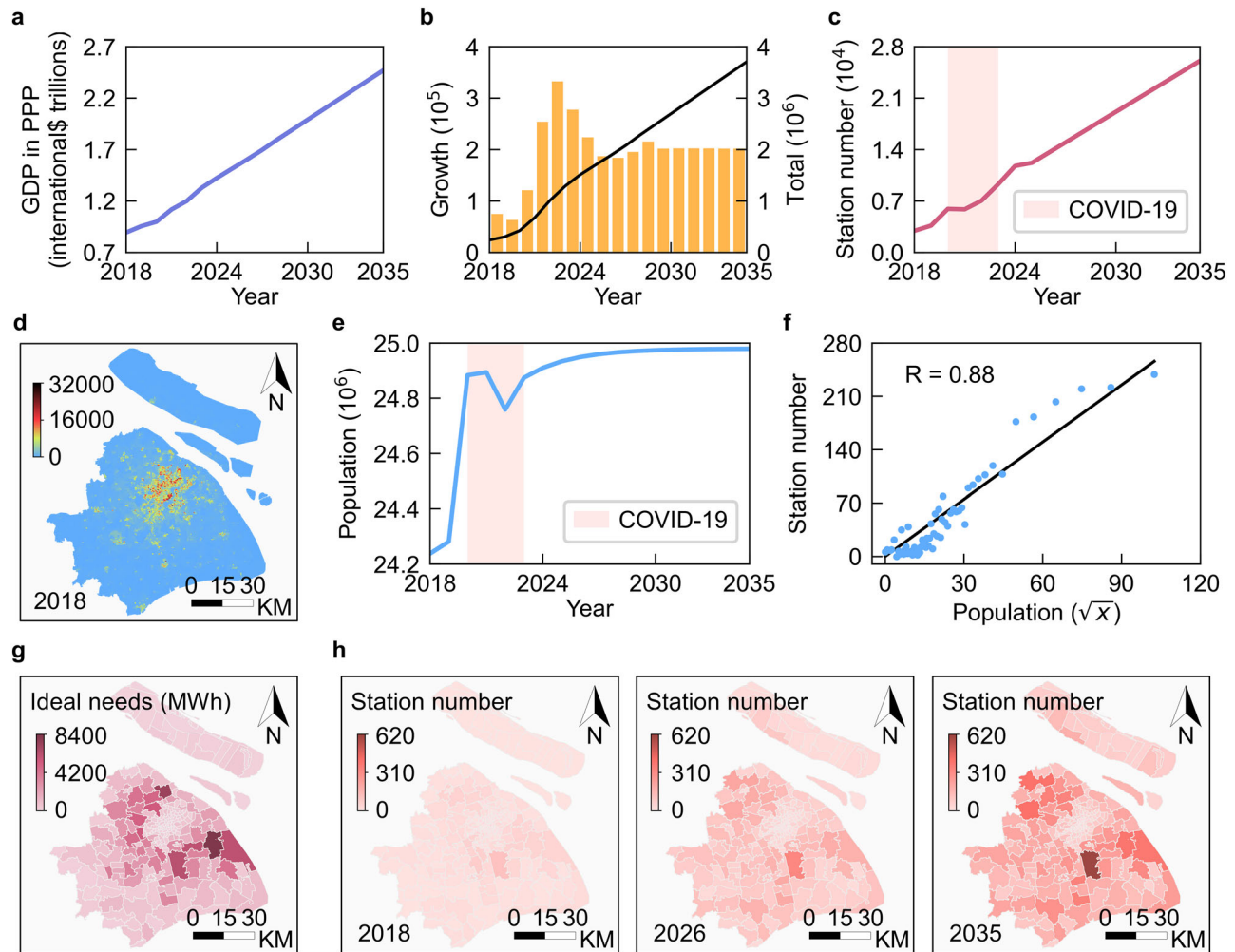


Fig. 4 | Projected GDP, population, EVs, charging station number, and spatial distributions by 2035 in Shanghai. **a** GDP in Purchasing Power Parity (PPP) terms estimation. **b** Annual EVs growth (bar) and ownership (curve) estimation. **c** Charging station number estimation. **d** Population of Shanghai in 2018. **e** Population of Shanghai from 2018 to 2035. **f** Correlation between the square root of

the regional population and the number of charging stations in the regions. Each region is comprised of 400 GIS cells with similar population density. **g** Charging demand after optimal scheduling. **h** Charging stations layout in 2018, 2026, and 2035, considering optimal scheduling and population growth.

increase in the coming years, leading to emissions disequilibrium across regions. Implementing effective charging scheduling and optimizing the placement of charging stations can greatly reduce the adverse impacts of charging on both the grid and the environment.

Sensitivity analysis of adoption rate and scheduling threshold

The effectiveness of the proposed scheduling strategy is dependent on driver adoption. While our preceding analysis assumed a 100% adoption rate to establish an idealized performance benchmark, this assumption does not hold in practice. To investigate the impact of partial compliance, we simulated the strategy's outcomes under different adoption rates (20%, 40%, 60%, 80%, and 100%), as detailed in Fig. 7. The adoption of the scheduling strategy is modeled deterministically to reflect a behaviorally plausible scenario where drivers who are least inconvenienced are most likely to participate. For each scenario, we first determine the optimal charging schedule for the entire EV fleet (i.e., 100% adoption). We then calculate the total scheduled time for each vehicle, which is the cumulative temporal shift of all its charging sessions compared to its original plan. All vehicles are then ranked in ascending order based on this total scheduled time. A given adoption rate is then applied by assuming that only the corresponding percentage of vehicles with the smallest total scheduled time will comply with the new schedule. This prioritized approach avoids the potential for suboptimal outcomes that

could arise from random adoption decisions and ensures that our analysis models a progressive uptake of the strategy. Our findings reveal a complex relationship between adoption rates and performance metrics. With full adoption, the peak power consumption reduction is maximized in 2022 at 390.97 kWh per vehicle per year (Fig. 7a), which is consistent with the peak preventable proportion observed in 2022 (Fig. 6b). However, under all partial adoption scenarios (20% to 80%), the greatest consumption reduction shifts to 2019, yielding savings of 5.85, 44.85, 119.75, and 219.45 kWh per vehicle per year, respectively. Notably, the maximum peak load reduction consistently occurs in 2019, regardless of the adoption rate (Fig. 7b).

The impact of the adoption rate on scheduling effectiveness is markedly non-linear. This phenomenon arises because our adoption model prioritizes participation from drivers who are least impacted by the scheduling. Consequently, at low adoption rates, the participating vehicles are those requiring only minor schedule adjustments, which have a correspondingly small effect on overall peak reduction. The disproportionately larger benefits are realized at higher adoption rates when vehicles requiring more substantial and thus more impactful charging shifts are included. This is evident in the load profiles (Fig. 7c), where the load-shifting effect at 60% adoption is substantially less than half of that at 100% adoption. The full adoption scenario clearly demonstrates a large transfer of energy demand from peak to off-peak hours.

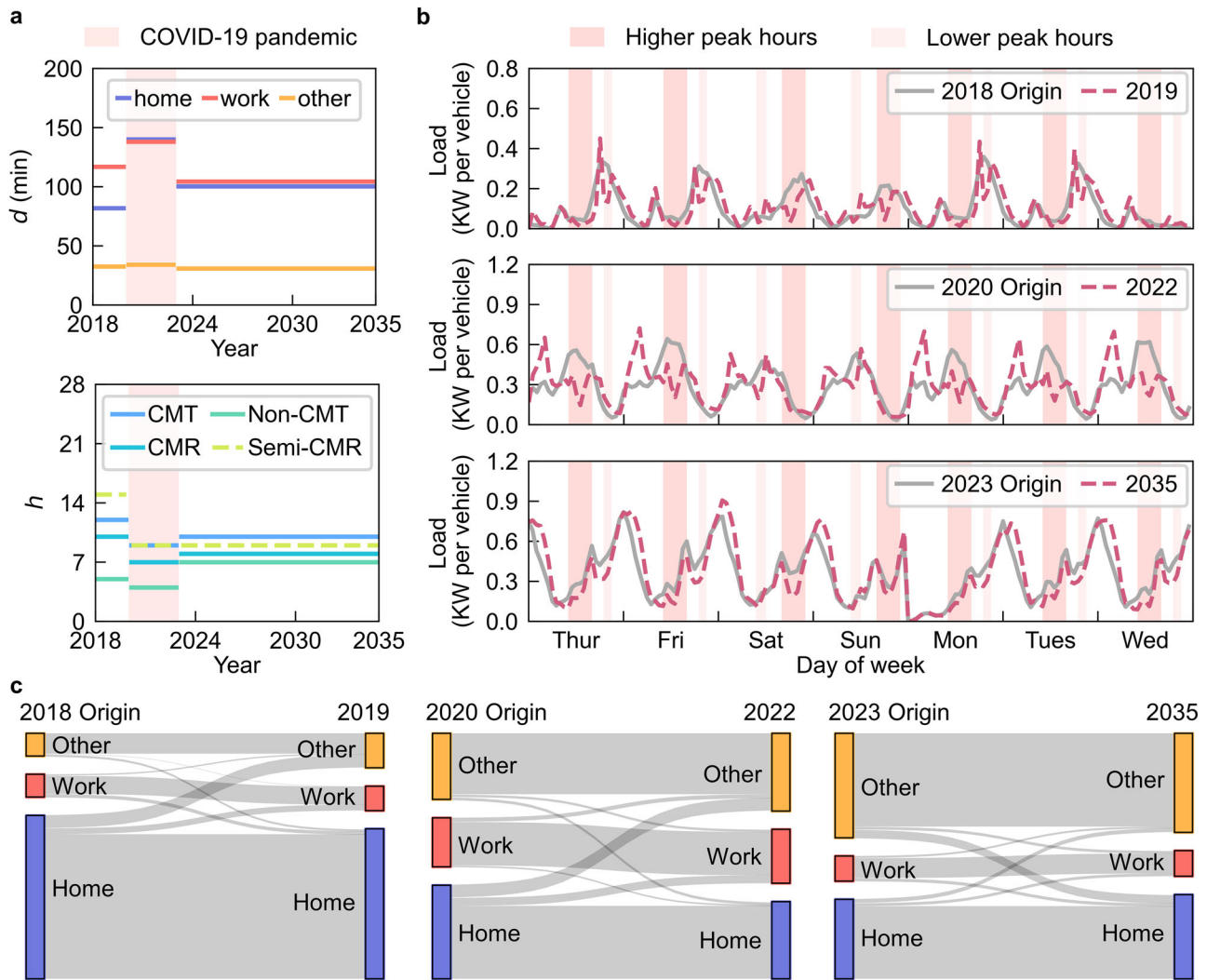


Fig. 5 | Impact of charging scheduling on charging demand after charging station layout. **a** Scheduling thresholds for pre-pandemic, intra-pandemic, and post-pandemic periods (parking time and stay number). **b** Pre- and post-scheduling

charging loads during pre-pandemic, intra-pandemic, and post-pandemic periods under a static demand scenario. **c** Pre- and post-scheduling charging areas during pre-pandemic, intra-pandemic, and post-pandemic periods.

Spatially, the scheduling strategy induces different geographical charging patterns across the years. For example, the 2018 scheduling incentivizes charging in central Shanghai over suburban areas. By contrast, the 2035 scheduling reverses this pattern, redirecting vehicles towards the suburbs, especially at a 60% adoption rate (Fig. 7d). Furthermore, the spatial redistribution of load is asymmetrical. In 2035, a 60% adoption rate results in a maximum regional load decrease of 175.09 MWh, while the corresponding load increase in other regions is only 83.19 MWh. This suggests that the strategy promotes grid balancing even under partial implementation.

The total dispatching power exhibits a non-monotonic increase over time, with great deviations from 2019 to 2022, likely reflecting pandemic-related disruptions in charging behavior (Fig. 7e). While extra emissions peak in 2019 across all partial adoption scenarios, they are projected to decrease sharply after 2023. This rapid decline is attributable to the decarbonization of the power grid through the large-scale integration of renewable power and improvements in transmission efficiency (Fig. 7f).

In addition to the adoption rate, the parameter R exerts an important influence on scheduling outcomes. This parameter serves a dual function, acting as a threshold for both EVs scheduling and charging station deployment. In our preceding analysis, R was set to 10%, a value chosen as a trade-off to minimize the impact on EV drivers, satisfy infrastructure requirements, and reduce computational overhead. However, it is observed

that the potential for both scheduling within stays and scheduling between stays increases with R , exhibiting a particularly sharp rise as R approaches 100% from 90%. This phenomenon is attributed to the long-tail effect in the distribution of parking time and stays number (Fig. 8a–b).

Due to computational constraints that preclude an exhaustive analysis, we selected four representative vehicle archetypes to investigate the influence of R on scheduling performance (Fig. 8c). These archetypes represent vehicles with: (1) minimal charging times and few stays (CMT), (2) maximal charging times (CMR), (3) maximal number of stays (Non-CMT), and (4) minimal charging times despite a maximal number of stays (Semi-CMR). The results indicate that the nature of scheduling varies greatly with R . The CMT rarely undergoes scheduling between stays owing to its limited number of stays. The CMR is rarely scheduled at all, a counter-intuitive finding given its frequent stays and high charging times. This is because its high operational uptime results in short dwell times and consistently low state-of-charge (SOC) at each stop, rendering it ineligible for either scheduling strategy. In contrast, the Non-CMT and Semi-CMR are amenable to both scheduling within stays and scheduling between stays (Fig. 8d–e).

Regarding the total scheduled time (Fig. 8f), all types of EVs except CMR (which has negligible scheduling potential) show an increase in scheduled time at higher R values. The behaviors of CMT and Non-CMT are particularly noteworthy. The CMT exhibits a substantial total scheduled time even at low R values, as it primarily engages in scheduling between

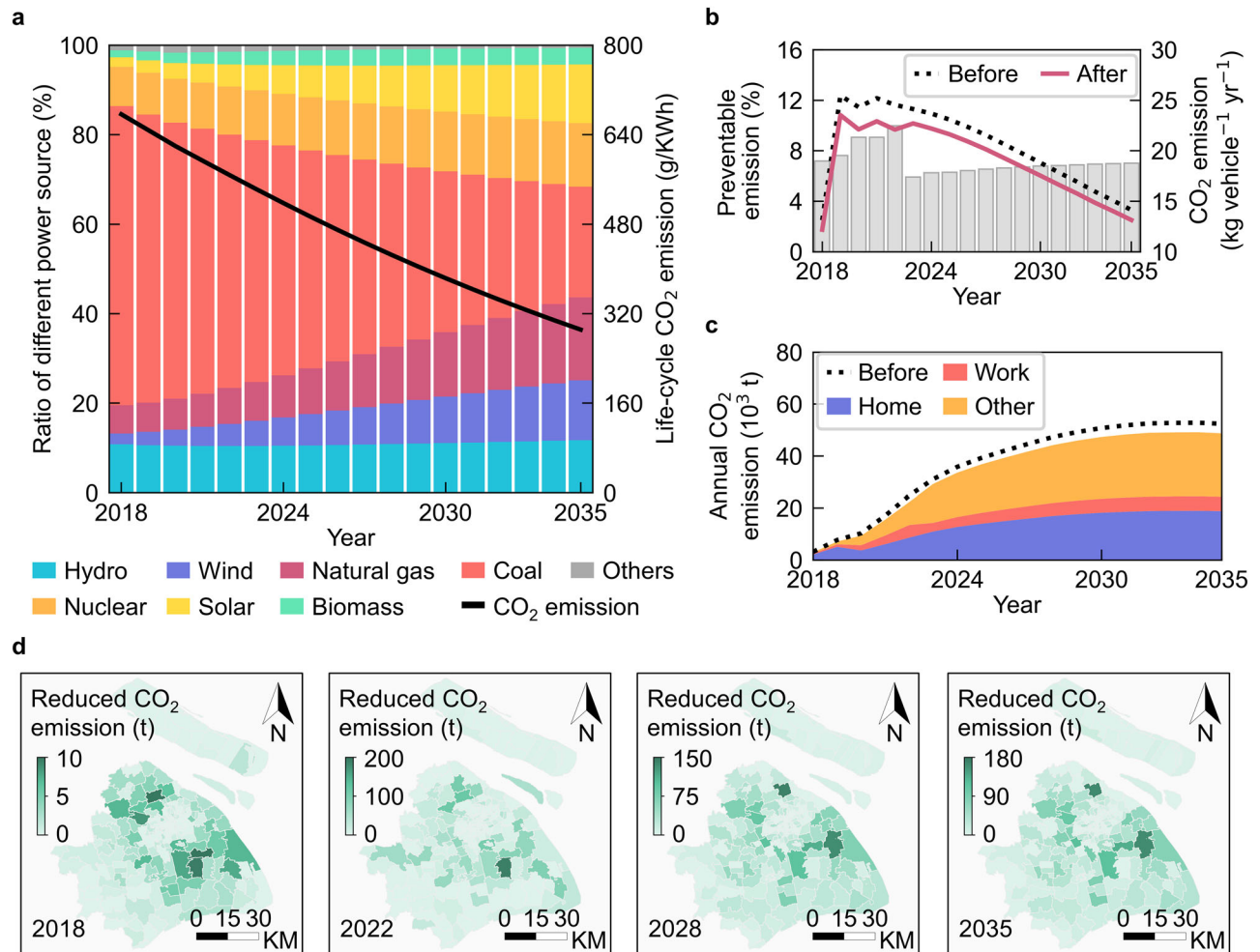


Fig. 6 | Prediction of power mix and potential grid emission reductions by applying scheduling. **a** The power mix (bar) and Life-cycle CO₂ emissions (curve) in East China. **b** Predicted emissions (line) and CO₂ preventable proportions (bar).

c Annual CO₂ emissions before and after scheduling. **d** Annual reduced CO₂ emissions space distribution after scheduling in 2018, 2022, 2028, and 2035.

stays, which involves a more large temporal shift of the charging session compared to scheduling within stays. However, the total scheduled time for the Non-CMT plateaus after R reaches 40%, indicating convergence to an optimal state. Finally, concerning the effectiveness in dispatching power reduction, the performance for all types of EVs, excluding the converged Non-CMT, improves monotonically with an increasing R .

Discussion

We develop a flexible charging scheduling strategy and charging station deployment policy based on the EV charging demand and urban mobility, analyzing their impacts on the power grid and associated CO₂ emissions. While the analysis for past years (2018–2025) serves as a retrospective benchmark under ideal conditions, it validates the strategy's robustness across diverse real-world mobility patterns. Our results show that the scheduling strategy can reduce the average daily peak power consumption by 97.12 MWh in 2018. Following the deployment of charging stations, the scheduling strategy can reduce the peak load by 446.79 MW, daily peak power consumption by 3.64 GWh, and annual CO₂ emissions from scheduling-related power losses by 3.68 thousand tons in 2035. From 2018 to 2035, the cumulative reductions in peak power consumption and CO₂ emissions are 10,608.82 GWh and 46.06 thousand tons, respectively. Reducing peak power consumption through scheduling can result in huge savings by reducing the need for peak plant deployment, which would otherwise cost several hundred million dollars (from hundreds to thousands of US dollars per kilowatt)⁴⁶. Additionally, it can decrease tens of millions of

dollars typically spent on capturing the indirect CO₂ emissions associated with dispatching peak power (from hundreds to a thousand US dollars per ton of CO₂)⁴⁷. Given that the construction of peak power plants and associated transmission infrastructures would result in increased environmental pollution and CO₂ emissions, the overall impact on cities would be tremendous. This highlights the urgent need for measures to encourage EV drivers to adopt more organized and grid-friendly charging behavior.

Our model adopts a conservative approach to the following parameters to ensure robust results, while the actual scheduling outcomes may be more favorable. Firstly, for the scheduling threshold, we select the decile distribution of vehicle mobility patterns to minimize the impact of scheduling. However, since most parking durations largely exceed our scheduling threshold (private vehicles may remain parked at home for eight hours), more effective, personalized scheduling strategies could be applied. The sensitivity analysis results demonstrate that our proposed scheduling strategy remains effective across a range of driver compliance levels and scheduling thresholds, underscoring the robustness and validity of our approach. Secondly, the deployment of charging stations adopts a traditional demand-driven approach, with fewer than 10% allocated for scheduling purposes. Consequently, deploying charging stations does not enable the optimal scheduling strategy by 2035. Finally, considering that the Chinese government may increase EV sales to 100% by 2035², and that Shanghai is one of the most economically developed and electrified regions in China, the actual number of EVs may largely exceed our estimation for the scheduling strategy.

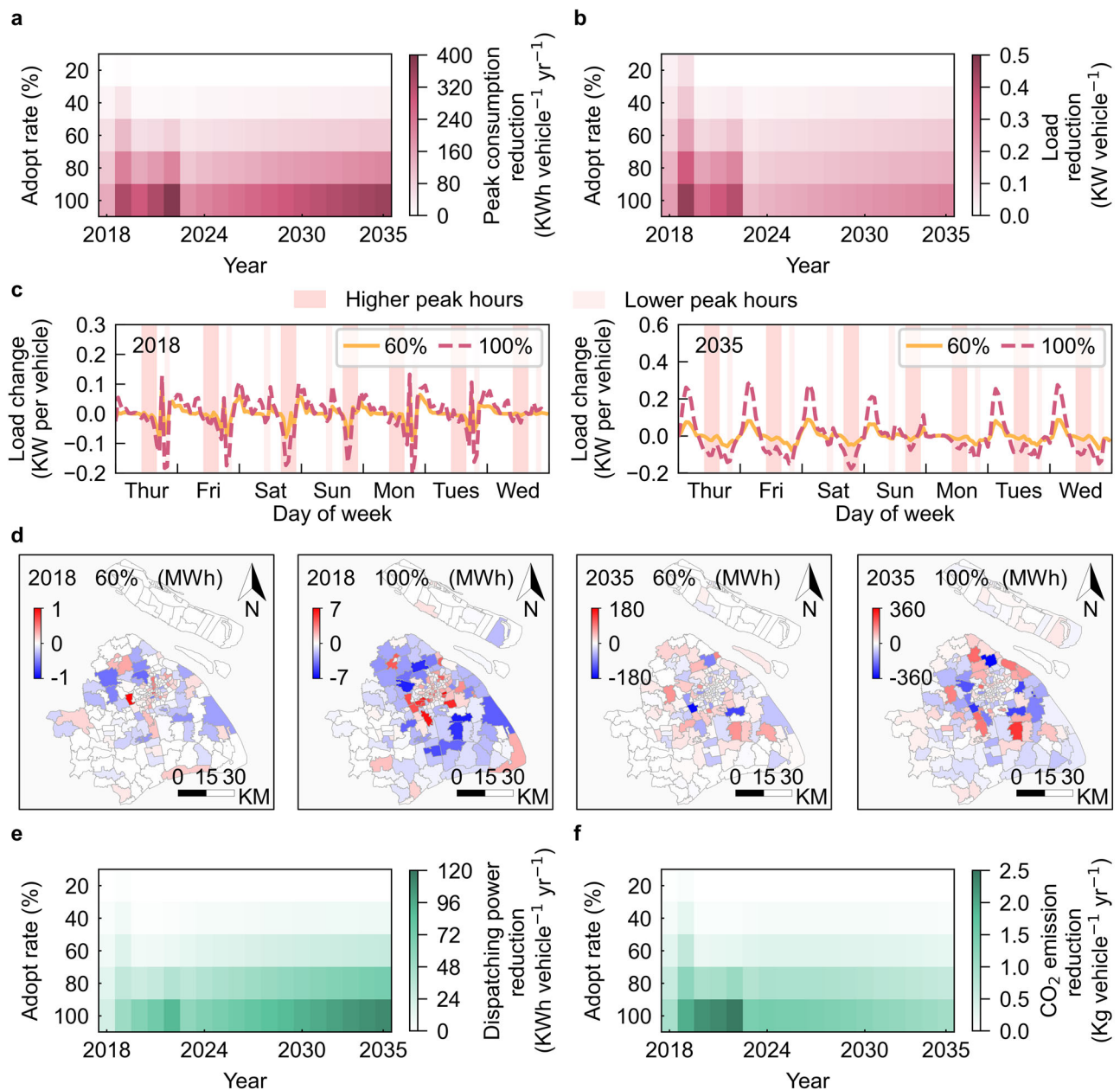


Fig. 7 | Impact of the scheduling strategy on charging behaviors and extra emissions under different adoption rates and years. a Peak hours power consumption reduction. **b** Maximum reduction during peak hour load. **c** Changes in

total charging loads at different adoption rates in 2018 and 2035. **d** Changes in the spatiotemporal distribution of charging demand at different adoption rates in 2018 and 2035. **e** Dispatching power consumption reduction. **f** Extra emission reduction.

Our analysis has several limitations that suggest avenues for future research. We assume that all drivers would comply with the scheduling requirements. However, drivers may opt out of scheduling due to cost, accessibility, or personal preferences. Therefore, future research should investigate the extent to which drivers accept scheduling strategies. To promote the adoption of scheduling strategies, policymakers or grid operators need to provide appropriate dynamic pricing costs and/or subsidies for charging cost reduction. Moreover, the lack of communication capabilities in EVs and the inability to regulate charging power are the main factors hindering smart charging development. To address this issue, vehicle drivers can communicate with the cloud through artificial intelligence driven applications to enable scheduling. EVs can continue charging at a minimal power level to address the issue that most EVs are unable to delay charging⁴⁸. Furthermore, this study does not model detailed grid dynamics such as voltage fluctuations or the specific technological heterogeneity of the EV fleet (e.g., varying battery capacities and charging rates). Such an analysis

would require granular grid topology and vehicle-specific data beyond the scope of our current work.

When estimating scheduling results, we don't account for the challenges associated with developing charging station communication systems and their potential impacts. The charging infrastructure's operational system needs to monitor grid status, renewable power generation, and EV charging in real time, while maintaining communication with both EVs and grid through cloud⁴⁹. The charging time of EVs, especially fast charging (from a few minutes to several hours), requires charging facilities to have highly automated control and high communication frequencies, which results in additional power load. It may raise privacy and regulatory concerns, as communication data includes the personal and travel information of EV drivers. Therefore, designing low-cost, energy-efficient, secure, and effective communication systems for charging infrastructure is essential. Furthermore, our model does not explicitly incorporate detailed transmission and distribution grid constraints. Shanghai is a consistent net importer

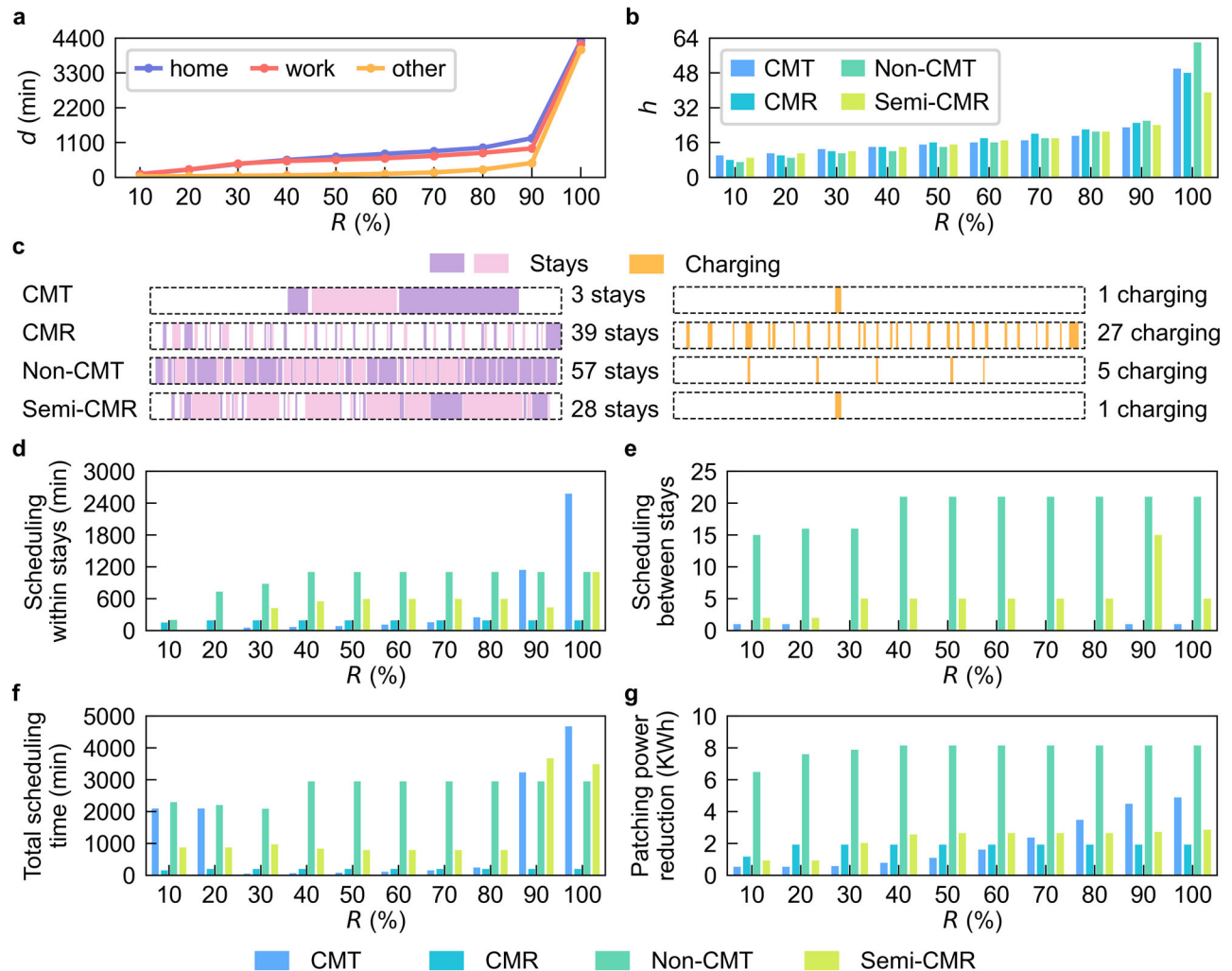


Fig. 8 | Scheduling thresholds and impacts of different R parameter values.

a Scheduling thresholds of scheduling within stays. **b** Scheduling thresholds of scheduling between stays. **c** Stays and charging event counts for four sample EVs. Note that colors are used only to visually distinguish consecutive stays and do not

represent stay type. **d** Magnitude of scheduling within stays. **e** Magnitude of scheduling between stays. **f** Magnitude of total scheduling. **g** Reduction in dispatching power consumption.

of electricity, meaning its peak demand is met by dispatching power from the wider East China power grid. This regional grid is heavily dominated by thermal power generation (approximately 70% in 2020). Consequently, in this context, any increase in peak load directly translates to increased fossil fuel consumption, making peak-shaving a primary and valid objective for both grid stability and decarbonization. Future research, particularly in regions with high renewable penetration where absorbing surplus generation is key, should indeed incorporate detailed grid constraints and potentially different optimization objectives.

Our study is based on observations in Shanghai, a global metropolis with one of the largest EV fleets in the world and ready access to high-resolution trajectory data. While testing on diverse urban datasets is a desirable future direction, Shanghai serves as a relevant and representative case study. As one of the world's leading cities in EV adoption, it provides a "stress-test" environment for grid integration, and the solutions proven here offer a strong proof-of-concept for other major cities poised for similar growth. However, the proposed strategies and findings are applicable to other regions in China and worldwide. Countries and regions like the United States, Japan, and the European Union are also experiencing rapid EV adoption, which could greatly strain the power grid's stability due to increased charging demand. While the issue can be addressed by deploying additional peak power plants and promoting renewable energy, more effective charging station deployment and scheduling strategies can reduce

peak power demand and mitigate the impact of EV growth on the power grid. This would support the sustainable development of both EVs and renewable energy. In conclusion, our research highlights the huge impacts of EV charging demand and scheduling strategies on the power system. It illustrates that coordinated policies are essential for managing the effects of charging on grid stability and associated emissions, while also advancing EV adoption. These measures will contribute to decarbonization in the transportation and power sectors, offering substantial environmental and public health benefits.

Methods

Data sources

Vehicle trajectory data have become indispensable for understanding mobility patterns and optimizing urban systems^{50–52}, serving as a cornerstone of digital city research. In this study, we utilize a high-resolution dataset consisting of continuous trajectory data collected from randomly selected EVs operating in Shanghai. The dataset collection spanned several sampling periods across 4 years: October 18–24, 2018 (6460 vehicles); August 1–29, October 15–November 14, and December 15–29, 2022 (6066 vehicles); January 1–14, 2023 (1179 vehicles); and July 15–21, 2024 (12,483 vehicles). Each EV within the dataset is tracked over a continuous one-week period. The 2023 data are primarily used as a validation set to reflect the characteristics of the pandemic era, given that data collection spanned the

official termination of China's COVID-19 emergency response on January 8, 2023.

The sample includes battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs). Each record includes anonymized vehicle IDs, timestamps, instantaneous speed, odometer readings, GPS coordinates, charging status, and state-of-charge (SOC) data. To ensure data integrity, we apply rigorous preprocessing procedures: invalid records are removed, missing values are either imputed through spatiotemporal interpolation or excluded, and trajectory drift artifacts are corrected using smoothing algorithms or discarded if they cannot be resolved.

For stop-type classification of EVs, this study employs the existing framework from ref. 36, wherein all vehicle stops are classified into three distinct categories: residential (home), occupational (workplace), and other locations.

The charging infrastructure data in this study comprise two categories: residential charging piles and public charging stations. In accordance with Shanghai's policy requiring home charging pile installation for all registered EVs, we assume that charging facilities are available at all vehicle stops identified as "residential". Public charging station data in 2018 are obtained through web scraping from an official platform⁵³, which comprise the following fields: operator and station identification codes, geospatial coordinates (latitude and longitude), counts of direct current piles (both total and available), counts of alternating current piles (both total and available), as well as electricity costs, service charges, and parking fees. Additionally, we collected data of public charging stations over 7 years (2018–2024). This dataset was subsequently used to forecast the proliferation of public charging stations. Demographic and economic data are obtained from the Shanghai Statistical Yearbook and Economic Census Database⁵⁴. Statistics on vehicle ownership, registrations, and retirements are sourced from the Shanghai Electric Vehicle Public Data Collecting, Monitoring, and Research Center in collaboration with municipal transportation authorities^{55,56}. Energy infrastructure data—encompassing power grid structure, generation portfolio, and associated emission metrics—are provided by the State Power Grid Corporation of China. These multidisciplinary datasets form the foundation for our scenario projections. Geographic zoning information is derived from shapefiles provided by the Shanghai Planning and Natural Resources Bureau⁵⁷.

Flexible charging scheduling to minimize exposure to grid peak

We proposed flexible charging scheduling based on the drivers' driving and charging patterns. To minimize the impact on drivers, we don't change the given travel route or arrival and departure time, but only schedule the charging locations and time based on the existing route. We also ensure the charging requirements are satisfied and limit the scheduling to an acceptable scope.

We assume that the grid load is $G(t)$, the maximum power generation capacity is G_m , and the charging power of the charging station is P . The drivers drive the EVs with the power consumption per unit distance as E and have n stays and m charging sessions. The drivers stay at $\{s_1, s_2, \dots, s_n\}$ with vehicle kilometers traveled (VKT) $\{x_{s_1}, x_{s_2}, \dots, x_{s_n}\}$ and charge at stays $\{s_{i_1}, s_{i_2}, \dots, s_{i_m}\}$, $i_k \in [1, n]$, $k \in [1, m]$. The EV drivers arrive at s_{i_k} at $t_a^{s_{i_k}}$ with SOC $\text{soc}_a^{s_{i_k}}$ and begin charging at $t_b^{s_{i_k}}$. They end up charging at $t_e^{s_{i_k}}$ and departure at $t_d^{s_{i_k}}$ with SOC $\text{soc}_d^{s_{i_k}}$. We schedule charging sessions for new stays $\{s_{j_1}, s_{j_2}, \dots, s_{j_m}\}$, $j_k \in [1, n]$, $k \in [1, m]$, according to the following rules:

- (1) Consistent with given driving patterns: the drivers' stay $\{s_1, s_2, \dots, s_n\}$, arrival time $\{t_a^{s_1}, t_a^{s_2}, \dots, t_a^{s_n}\}$, and departure time $\{t_d^{s_1}, t_d^{s_2}, \dots, t_d^{s_n}\}$ keep constant. The scheduling can only alter the charging location and time within the n -th stay.
- (2) Consistent with given charging patterns: the drivers' charging session number m , SOC at the beginning $\text{soc}_a^{s_1}$, and SOC at the end $\text{soc}_d^{s_n}$ keeps

constant. After charging session s_{i_k} scheduling to s_{j_k} , the end SOC of charging sessions, except for the last one after scheduling remains consistent, i.e., $\text{soc}_d^{s_{i_k}} = \text{soc}_d^{s_{j_k}}$, $k \in [1, m-1]$.

- (3) Scheduling charging sessions between stays: The drivers can move the charging session $\{s_{i_1}, s_{i_2}, \dots, s_{i_m}\}$ forward or backward to $\{s_{j_1}, s_{j_2}, \dots, s_{j_m}\}$ within the threshold h , i.e., $|j_k - i_k| < h$, $k \in [1, m]$. We assume that the maximum proportion of scheduling available for scheduling is R (set to 10% in this study). It means we adopt the decile of the daily stay frequency as h to minimize the impact on drivers. This conservative choice ensures the required scheduling flexibility is limited to a level already common to 90% of drivers. Specifically, the value is set to 2 for CMT, 1 for CMR, 1 for Non-CMT, and 2 for Semi-CMR before the pandemic (2019 and earlier), 1 for CMT, 1 for CMR, 1 for Non-CMT, and 1 for Semi-CMR during the pandemic (2020–2022) and after the pandemic (2023 and onward), see Fig. 5a. Notably, Fig. 1e and Fig. 5a displays the weekly stay frequency distribution. Therefore, the daily stay threshold is derived by dividing by seven and then rounding to the nearest integer.

- (4) Guaranteeing travel demands: After charging session i_k is scheduled to j_k , SOC at arrival is updated as follows:

$$\text{soc}_a^{s_{j_k}} = \begin{cases} \text{soc}_a^{s_{i_1}}, k = 1, j_1 = 1 \\ \text{soc}_d^{s_{i_1}} - E \cdot (x_{s_{j_k}} - x_1), k = 1, j_1 \neq 1 \\ \text{soc}_d^{s_{i_{k-1}}} - E \cdot (x_{s_{j_k}} - x_{s_{j_{k-1}}}), k \neq 1 \end{cases} \quad (1)$$

where $k \in [1, m]$, $\text{soc}_a^{s_{j_k}} \geq 0$. If the final charging session $s_{i_m} \neq s_{j_m}$, $\text{soc}_d^{s_{j_m}}$ can be changed to keep $\text{soc}_d^{s_n}$ constant. We require $\text{soc}_d^{s_{j_m}} \leq 100$.

- (5) Updating charging time: We assume that charging starts immediately after the drivers come to a stay, i.e., $t_b^{s_{i_k}} = t_a^{s_{i_k}}$. After charging session s_{i_k} is scheduled to s_{j_k} , charging beginning time and charging end time is updated as follows:

$$t_b^{s_{j_k}} = t_a^{s_{j_k}} \quad (2)$$

$$t_e^{s_{j_k}} = t_a^{s_{j_k}} + \frac{(\text{soc}_d^{s_{j_k}} - \text{soc}_a^{s_{j_k}}) \cdot E}{P} \quad (3)$$

where we require $t_e^{s_{j_k}} \leq t_d^{s_{j_k}}$.

- (6) Scheduling charging sessions within stays: The charging beginning and end times at stay s_{j_k} can be delayed within threshold d . It can be represented as follows:

$$t_b^{s_{j_k}'} = t_b^{s_{j_k}} + d^{s_{j_k}} \quad (4)$$

$$t_e^{s_{j_k}'} = t_e^{s_{j_k}} + d^{s_{j_k}} \quad (5)$$

where we require $d^{s_{j_k}} \leq \min\{d, t_d^{s_{j_k}} - t_e^{s_{j_k}}\}$, which means any delayed charging session is strictly constrained to finish before the original departure time $t_d^{s_{j_k}'} \leq t_d^{s_{j_k}}$. We assume that the maximum proportion of scheduling available for scheduling is R (set to 10% in this study). It means we adopt the decile of the parking time as d , serving as a strict constraint to minimize driver inconvenience. Specifically, the value is set to 81.83 for Home, 116.82 for Work, and 32.5 for Other before pandemic (2019 and earlier), 140.00 for Home, 138.00 for Work, and 34.00 for Other during the pandemic (2020–2022), 100.28 for Home, 104.33 for Work, and 30.82 for Other after the pandemic (2023 and onward) (Fig. 5a).

- (7) Minimize the dispatch of electricity: Based on the above rules, we schedule the new charging sessions $\{s_{j_1}, s_{j_2}, \dots, s_{j_m}\}$. As well as the

corresponding charging beginning and end time $\{t_b^{s_{1'}}, t_b^{s_{2'}}, \dots, t_b^{s_{m'}}\}$ and $\{t_e^{s_{1'}}, t_e^{s_{2'}}, \dots, t_e^{s_{m'}}\}$. To quantify the impact of scheduling on the power grid, we choose the minimization of electricity dispatch as the ultimate objective. Solving for the minimum dispatch electricity utilization with the following equation:

$$\min D = \sum_{c=1}^k \int_{t_b^{s_{c'}}}^{t_e^{s_{c'}}} \left(\frac{G(t) - G_m}{G(t)} \right) \cdot P \quad (6)$$

Charging stations layout

To balance scheduling requirements with charging demands, we develop a set of rules based on population growth and scheduling objectives. We split the study area into 0.5 km square cells with a total number N . The cells are arranged in order of population size, denoted as $\{g_1, g_2, \dots, g_N\}$, where each cell contains population $\{p_1, p_2, \dots, p_N\}$, and the charging stations number $\{c_1, c_2, \dots, c_N\}$.

(1) Cell aggregation. Given the discreteness in population and charging station distribution across cells, we aggregate n (where $n = 400$ in this study) neighboring cells with similar population sizes to form a new region for further analysis. Following aggregation, the data comprises M distinct regions $\{r_1, r_2, \dots, r_M\}$. The distinct regions r_i can be represented as follows:

$$r_i = \{g_{ni-(n-1)}, g_{ni-(n-2)}, \dots, g_{\min(ni, N)}\} \quad (7)$$

where $i < \frac{N}{n} + 1$. The total population of new region is $\{P_1, P_2, \dots, P_M\}$, and the total charging stations is $\{C_1, C_2, \dots, C_M\}$. The P_i and C_i are calculated with the following equation:

$$P_i = \sum_{j=ni-(n-1)}^{\min(ni, N)} p_j \quad (8)$$

$$C_i = \sum_{j=ni-(n-1)}^{\min(ni, N)} c_j \quad (9)$$

(2) Allocation of charging station growth to regions. The total growth of charging stations is G . The allocation of charging stations allocation $\{G_1, G_2, \dots, G_M\}$ to region $\{r_1, r_2, \dots, r_M\}$ is determined by conditions as follows:

$$\frac{G_1}{\sqrt{P_1}} = \frac{G_2}{\sqrt{P_2}} = \dots = \frac{G_M}{\sqrt{P_M}} \quad (10)$$

where $\sum_{i=1}^M G_i = G$. To ensure that each region is allocated at least one charging station, we set $G_i \geq 1, i \in [1, M]$.

(3) Allocation of charging station growth to cells based on scheduling. We assume that the maximum proportion of charging stations available for scheduling is R (set to 10% in this study). This parameter is intentionally aligned with the 10th percentile threshold used for our behavioral scheduling constraints, reflecting a unified conservative modeling philosophy. Just as we limit required driver flexibility on the demand side, we similarly limit reliance on specialized infrastructure on the supply side. This ensures the 90% of new stations serve organic demand, while a smaller, strategic portion enables grid-balancing services, making our results robust and practical. For region r_k , where the charging station growth is G_k , the maximum number of charging stations available for scheduling is RG_k . Region r_k contains cells $\{g_{nk-(n-1)}, g_{nk-(n-2)}, \dots, g_{\min(nk, N)}\}$, and there are s cells that require

the deployment of charging stations for scheduling purposes. we randomly select G_d cells to deploy charging stations within the s cells, leaving G_r stations remaining for deployment. The number of charging stations deploying for scheduling G_d and number of remaining stations G_r can be calculated as follows:

$$G_d = \begin{cases} RG_k, & s \geq RG_k \\ s, & s < RG_k \end{cases} \quad (11)$$

$$G_r = \begin{cases} G_r = (1 - R)G_k, & s \geq RG_k \\ G_k - s, & s < RG_k \end{cases} \quad (12)$$

(4) Allocation of remaining charging stations based on population growth. If $G_r \geq \min(nk, N) - n(k - 1)$, we deploy a charging station in each cell, and repeat this process until the remaining charging stations are less than $\min(nk, N) - n(k - 1)$. If $G_r < \min(nk, N) - n(k - 1)$, we calculate the per capita charging station as follows:

$$\text{PCCS}_{r_k} = \left\{ \frac{c_{nk-(n-1)}}{p_{nk-(n-1)}}, \frac{c_{nk-(n-2)}}{p_{nk-(n-2)}}, \dots, \frac{c_{\min(nk, N)}}{p_{\min(nk, N)}} \right\} \quad (13)$$

where PCCS_{r_k} is the per capita charging station in r_k , The results are then reordered in ascending order:

$$\text{PCCS}_{r_k}^{\text{ascending}} = \left\{ \frac{c_{k_1}}{p_{k_1}}, \frac{c_{k_2}}{p_{k_2}}, \dots, \frac{c_{k_{\min(nk, N) - n(k-1)}}}{p_{k_{\min(nk, N) - n(k-1)}}} \right\} \quad (14)$$

where $\frac{c_{k_1}}{p_{k_1}} < \frac{c_{k_2}}{p_{k_2}} < \dots < \frac{c_{k_{\min(nk, N) - n(k-1)}}}{p_{k_{\min(nk, N) - n(k-1)}}}$, $k_m \in [nk - (n - 1), \min(nk, N)]$.

Finally, we deploy one charging station at $\{g_{k_1}, g_{k_2}, \dots, g_{k_{G_r}}\}$.

Reporting summary

Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Grid load in Shanghai, power mix in East China, results of charging load and charging demand after scheduling are available via Figshare at <https://doi.org/10.6084/m9.figshare.30970267>⁵⁸. The vehicle trajectory data used in this study are available upon formal request to the Shanghai Electric Vehicle Public Data Collection, Monitoring, and Research Center. All supporting datasets (i.e., public charging station records, socioeconomic indicators, municipal vehicle statistics, power grid infrastructure data, and regional geospatial files) are accessible without restrictions.

Code availability

The code for GDP, EV fleet size, charging station numbers, and power mix estimation is available at the following GitHub repository: <https://github.com/Liaochenlei/Electric-Vehicle-Charging>.

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Author contributions

C.L. and X.M.C. proposed the question. C.L. and J.D. designed and conducted the experiments. C.L. and J.D. developed the algorithms. C.L., J.D., X.M.C. and Q.Y. wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Xiqun Michael Chen or Quan Yuan.

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